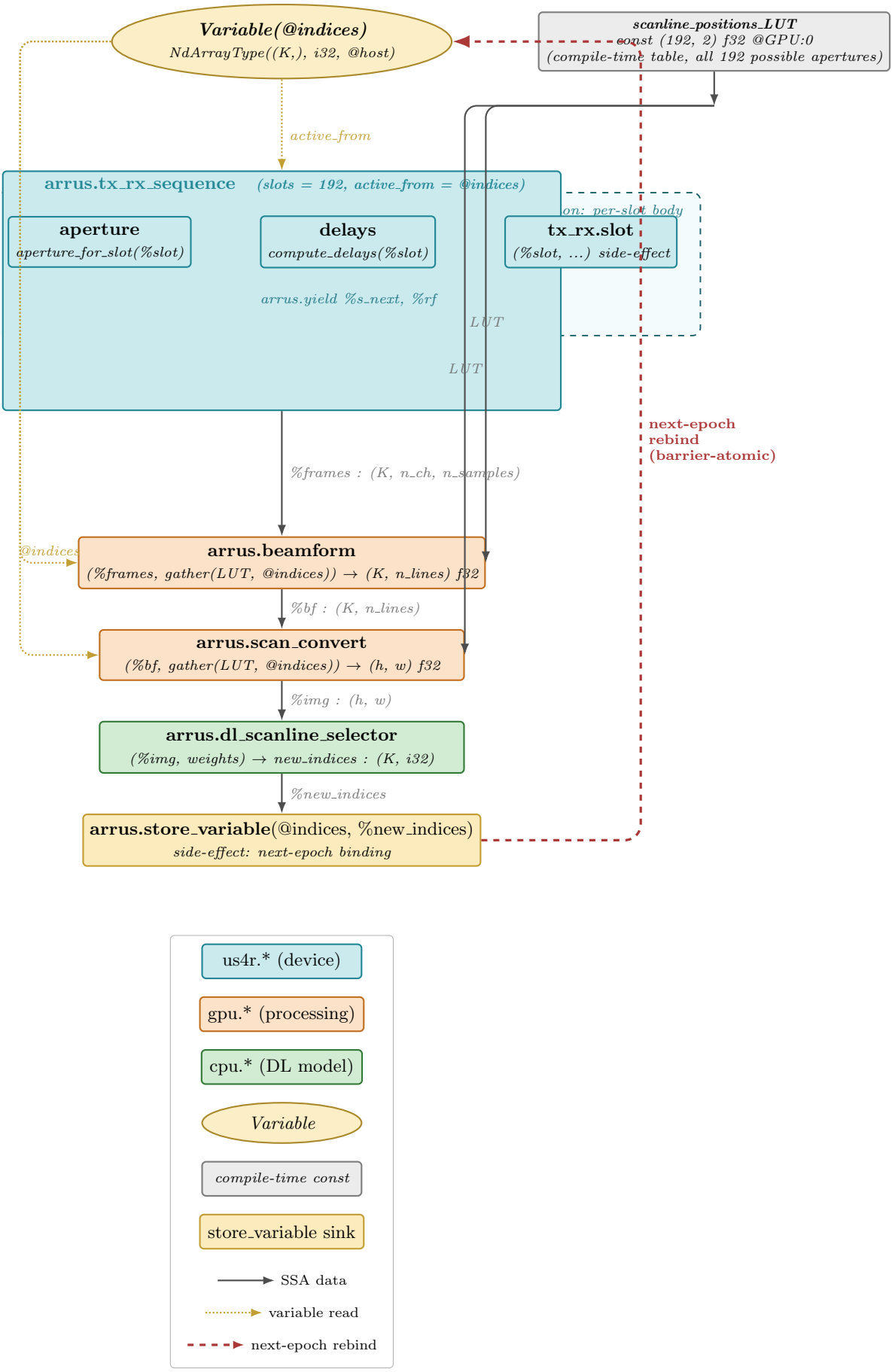

Variable(@indices)
NdArrayType((K,), i32, @host)
scanline_positions_LUT
const (192, 2) f32 @GPU:0
(compile-time table, all 192 possible apertures)
active_from
arrus.tx_rx_sequence (slots = 192, active_from = @indices)
on: per-slot body
aperture
aperture_for_slot(%slot)
delays
compute_delays(%slot)
tx_rx.slot
(%slot, ...) side-effect
arrus.yield %s_next, %rf
LUT
LUT
next-epoch
rebind
(barrier-atomic)
%frames : (K, n_ch, n_samples)
@indices
arrus.beamform
(%frames, gather(LUT, @indices)) → (K, n_lines) f32
%bf : (K, n_lines)
arrus.scan_convert
(%bf, gather(LUT, @indices)) → (h, w) f32
%img : (h, w)
arrus.dl_scanline_selector
(%img, weights) → new_indices : (K, i32)
%new_indices
arrus.store_variable(@indices, %new_indices)
side-effect: next-epoch binding
us4r.* (device)
gpu.* (processing)
cpu.* (DL model)
Variable
compile-time const
store_variable sink
SSA data
variable read
next-epoch rebind